

NMOS Model

LEVEL = 1	VTO = 0.7	GAMMA = 0.45	PHI = 0.9
NSUB = 9e+14	LD = 0.08e-6	UO = 350	LAMBDA = 0.1
TOX = 9e-9	PB = 0.9	CJ = 0.56e-3	CJSW = 0.35e-11
MJ = 0.45	MJSW = 0.2	CGDO = 0.4e-9	JS = 1.0e-8

PMOS Model

LEVEL = 1	VTO = -0.8	GAMMA = 0.4	PHI = 0.8
NSUB = 5e+14	LD = 0.09e-6	UO = 100	LAMBDA = 0.2
TOX = 9e-9	PB = 0.9	CJ = 0.94e-3	CJSW = 0.32e-11
MJ = 0.5	MJSW = 0.3	CGDO = 0.3e-9	JS = 0.5e-8

VTO: threshold voltage with zero V_{SB} (unit: V)

GAMMA: body-effect coefficient (unit: $V^{1/2}$)

PHI: $2\Phi_F$ (unit: V)

TOX: gate-oxide thickness (unit: m)

NSUB: substrate doping (unit: cm^{-3})

LD: source/drain side diffusion (unit: m)

UO: channel mobility (unit: $\text{cm}^2/\text{V/s}$)

LAMBDA: channel-length modulation coefficient (unit: V^{-1})

CJ: source/drain bottom-plate junction capacitance per unit area (unit: F/m^2)

CJSW: source/drain sidewall junction capacitance per unit length (unit: F/m)

PB: source/drain junction built-in potential (unit: V)

MJ: exponent in CJ equation (unitless)

MJSW: exponent in CJSW equation (unitless)

CGDO: gate-drain overlap capacitance per unit width (unit: F/m)

CGSO: gate-source overlap capacitance per unit width (unit: F/m)

JS: source/drain leakage current per unit area (unit: A/m^2)

Use $V_{DD} = 3 \text{ V}$ if not specified in problem

$$\epsilon_{\text{ox}} = 3.9$$

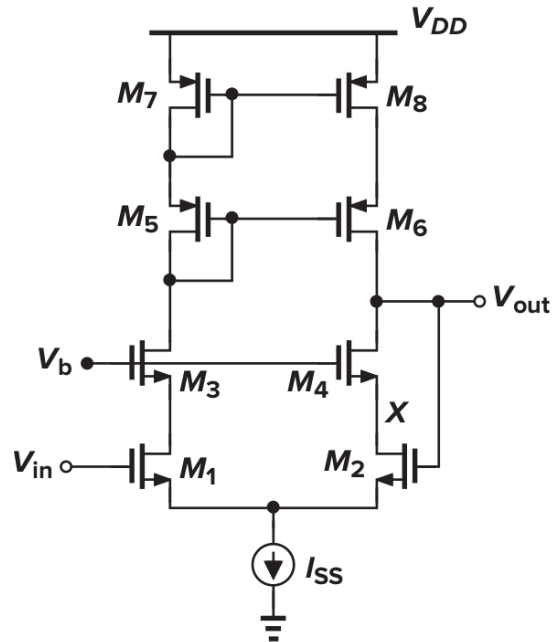
$$C_{\text{ox}} = 0.38 \text{ } \mu\text{m}$$

$$\mu_n C_{\text{ox}} = 134.28 \text{ } \mu\text{A/V}^2$$

$$\mu_p C_{\text{ox}} = 38.36 \text{ } \mu\text{A/V}^2$$

Ex 1: In the circuit of figure below, assume that $(W/L)_{1-4} = 100/0.5$, $I_{SS} = 1\text{mA}$, $V_b = 1.4\text{V}$, and $\gamma = 0$.

- If $M_5 - M_8$ are identical and have a length of $0.5\text{ }\mu\text{m}$, calculate their minimum width such that M_3 operates in saturation.
- Calculate the maximum output voltage swing.
- What is the open-loop voltage gain?



Answers:

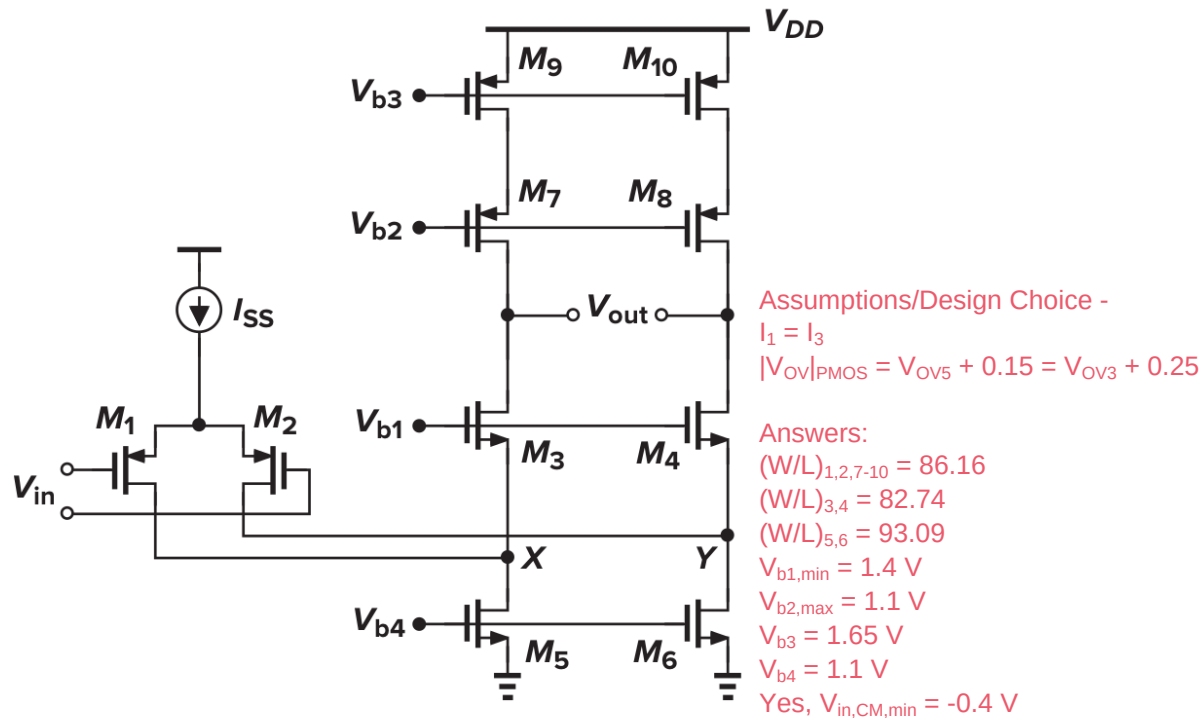
a) $(W/L)_{5-8, \min} = 212.77$

$W_{5-8, \min} = 106.38\text{ }\mu\text{m}$

b) swing = 0.507 V

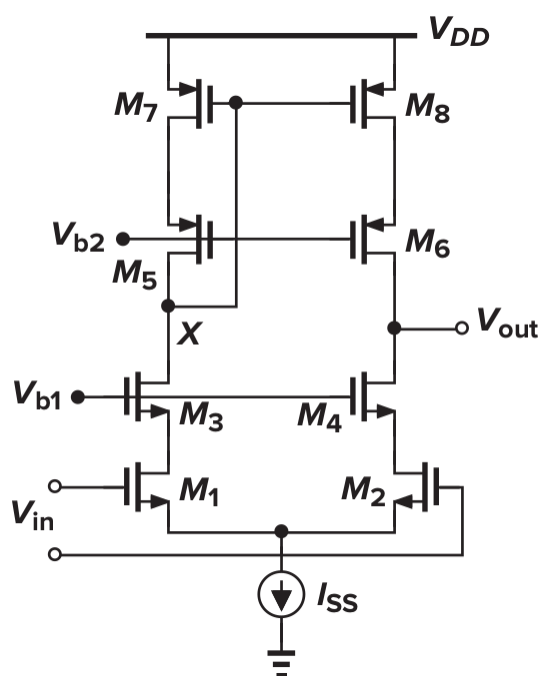
c) $A_v = 1301.32\text{ V/V}$

Ex 2: Design the folded-cascode op amp of figure below for the following requirements: maximum differential swing = 2.4 V, total power dissipation = 6 mW. If all of the transistors have a channel length of $0.5 \mu\text{m}$, what is the overall voltage gain? Can the input common-mode level be as low as zero?



Ex 3: In the op amp of figure below, $(W/L)_{1-8} = 100/0.5$, $I_{SS} = 1\text{mA}$, and $V_{b1} = 1.7\text{V}$. Assume that $\gamma = 0$.

- What is the maximum allowable input CM level?
- What is V_X ?
- What is the maximum allowable output swing if the gate of M_2 is connected to the output?
- What is the acceptable range of V_{b2} ?



- Answers:
 a) $V_{\text{in,CM,max}} = 1.507\text{V}$
 b) $V_X = 1.839 \text{ V}$
 c) swing = 0.507 V
 $1 \text{ V} < V_{\text{out}} < 1.507 \text{ V}$
 d) $1.039 \text{ V} < V_{b2} < 1.478 \text{ V}$